Final Exam

● Graded

Student

Emilia Girardin

Total Points

37 / 45 pts

Question 1

Overall Grades **37** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 8 pts** Point adjustment

MATH 202–001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
EMILIA GIRARDIN	eg230	Emilia Girardin

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: Emilia Girardin

IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	5	5	5	5	5	5	1.5	0.5	37

5/5

Problem 1. [5'] Find the general solution of the linear system:

$$2x - 3y = 1$$
$$-2y - 3z = 1$$
$$x + 2z = 1$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 0 & 1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right] \quad R_1 \leftarrow R_1 - 2R_3 \quad \left[\begin{array}{ccc|c} 0 & -3 & -4 & -1 \\ 0 & -2 & -3 & 1 \\ 1 & 0 & 2 & 1 \end{array}\right]$$

$$R_2 \leftarrow R_2 - R_1 \quad \left[\begin{array}{ccc|c} 0 & -3 & -4 & -1 \\ 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 \end{array}\right] \quad R_1 \leftarrow R_1 + 3R_2 \quad \left[\begin{array}{ccc|c} 0 & 0 & -1 & 5 \\ 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 \end{array}\right]$$

$$R_1 \leftrightarrow R_3 \quad R_3 \leftarrow -R_3 \quad \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & -5 \end{array}\right] \quad R_2 \leftarrow R_2 - R_3 \quad R_1 \leftarrow R_1 - 2R_3 \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -5 \end{array}\right]$$

General solution: $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ 7 \\ -5 \end{bmatrix}$

5/5

Problem 2. [5'] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

- ☒ a basis of the column space of A,
- ☒ a basis of the row space of A,
- ☒ and a basis of the null space of A.

$$\begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \quad R_2 \leftarrow \tfrac{1}{5}R_2 \quad R_1 \leftarrow R_1 - 2R_4 \quad \begin{bmatrix} 0 & -2 & -2 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$$

OK

$$R_1 \leftarrow R_1 + 2R_3 \quad R_4 \leftarrow R_4 + R_2 \quad \begin{bmatrix} 0 & 0 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 3 & 3 \end{bmatrix} \quad R_2 \leftarrow -R_2 \quad R_4 \leftarrow R_4 - 3R_3 \quad \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Basis of the row space: R_2, R_3: $\{[-5\ 0\ 5], [0\ 1\ 1]\}$

$$R_1 \leftrightarrow R_2 \quad R_2 \leftrightarrow R_3 \quad \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Basis of the column space: $\left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad z = t \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ -t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Basis of the null space: $\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

Problem 3. [5'] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

$v_1 = \begin{bmatrix}0\\1\\-1\end{bmatrix}$ $v_2 = \begin{bmatrix}1\\0\\-1\end{bmatrix}$ $v_3 = \begin{bmatrix}0\\0\\-1\end{bmatrix}$

Express e_1, e_2, e_3 as linear combinations of v_1, v_2, v_3

$$\left[\begin{array}{ccc|ccc}0&1&0&1&0&0\\1&0&0&0&1&0\\-1&-1&-1&0&0&1\end{array}\right] \quad R_3 \leftarrow R_3 + R_2 \quad \left[\begin{array}{ccc|ccc}0&1&0&1&0&0\\1&0&0&0&1&0\\0&-1&-1&0&1&1\end{array}\right]$$

$$R_3 \leftarrow R_3 + R_1 \quad \left[\begin{array}{ccc|ccc}0&1&0&1&0&0\\1&0&0&0&1&0\\0&0&-1&1&1&1\end{array}\right] \quad \begin{matrix}R_1 \leftrightarrow R_2\\R_3 \leftarrow -R_3\end{matrix} \quad \left[\begin{array}{ccc|ccc}1&0&0&0&1&0\\0&1&0&1&0&0\\0&0&1&-1&-1&-1\end{array}\right]$$

$e_1 = v_2 - v_3$, $T(e_1) = \begin{bmatrix}0\\2\end{bmatrix} - \begin{bmatrix}2\\2\end{bmatrix} = \begin{bmatrix}-2\\0\end{bmatrix}$

$e_2 = v_1 - v_3$, $T(e_2) = \begin{bmatrix}1\\1\end{bmatrix} - \begin{bmatrix}2\\2\end{bmatrix} = \begin{bmatrix}-1\\-1\end{bmatrix}$

$e_3 = -v_3$, $T(e_3) = -\begin{bmatrix}2\\2\end{bmatrix} = \begin{bmatrix}-2\\-2\end{bmatrix}$

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = T\left(x\begin{bmatrix}1\\0\\0\end{bmatrix} + y\begin{bmatrix}0\\1\\0\end{bmatrix} + z\begin{bmatrix}0\\0\\1\end{bmatrix}\right) =$$

$$= x\,T\left(\begin{bmatrix}1\\0\\0\end{bmatrix}\right) + y\,T\left(\begin{bmatrix}0\\1\\0\end{bmatrix}\right) + z\,T\left(\begin{bmatrix}0\\0\\1\end{bmatrix}\right) =$$

$$= x\begin{bmatrix}-2\\0\end{bmatrix} + y\begin{bmatrix}-1\\-1\end{bmatrix} + z\begin{bmatrix}-2\\-2\end{bmatrix} = \begin{bmatrix}-2&-1&-2\\0&-1&-2\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}$$

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = \begin{bmatrix}-2&-1&-2\\0&-1&-2\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}$$

5/5

Problem 4. [5'] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

$J = P^{-1} A P$

- $\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{vmatrix} = (1-\lambda)(2-\lambda)(1-\lambda) =$

$= (1-\lambda)^2(2-\lambda)$

$\lambda_1 = 1$ algebraic multiplicity = 2

$$A - I = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix} \Rightarrow \left[\begin{array}{ccc|c} 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & -2 & 0 & 0 \end{array}\right] \quad \begin{matrix} R_1 \leftarrow R_1 - 2R_2 \\ R_3 \leftarrow -\frac{1}{2} R_3 \end{matrix} \quad \left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{array}\right]$$

$$\begin{matrix} R_3 \leftarrow R_3 - R_2 \\ R_3 \leftrightarrow R_1 \end{matrix} \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad z = t,\ x = 0,\ y = 0 \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

geometric multiplicity = 1

Jordan block: $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ $\quad v_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$\lambda_2 = 2$ algebraic multiplicity = 1

$$A - 2I = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & -2 & -1 \end{bmatrix} \Rightarrow \left[\begin{array}{ccc|c} -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -2 & -2 & -1 & 0 \end{array}\right] \quad R_3 \leftarrow R_3 - 2R_1 \left[\begin{array}{ccc|c} -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -6 & -1 & 0 \end{array}\right]$$

$$\begin{matrix} R_2 \leftrightarrow R_3 \\ R_2 \leftarrow -\frac{1}{6} R_2 \\ R_1 \leftarrow -R_1 \end{matrix} \quad \left[\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 0 & 1 & 1/6 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad R_1 \leftarrow R_1 + 2R_2 \quad \left[\begin{array}{ccc|c} 1 & 0 & 2/6 & 0 \\ 0 & 1 & 1/6 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \begin{matrix} z = t \\ x = -\frac{2}{6} t \\ y = -\frac{1}{6} t \end{matrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2/6\, t \\ -1/6\, t \\ t \end{bmatrix} = t \begin{bmatrix} -2/6 \\ -1/6 \\ 1 \end{bmatrix} \quad \text{geometric multiplicity} = 1 \quad v_3 = \begin{bmatrix} -2/6 \\ -1/6 \\ 1 \end{bmatrix}$$

Jordan block: $[2]$ $\Rightarrow$ $J = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ $\quad P = \begin{bmatrix} 0 & v_{2x} & -2/6 \\ 0 & v_{2y} & -1/6 \\ 1 & v_{2z} & 1 \end{bmatrix}$

5/5

Problem 5. [5'] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$$A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}$$

$$\det(A-\lambda I) = \begin{vmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{vmatrix} = (3-\lambda)(3-\lambda)(4-\lambda) - \left[(3-\lambda) + (3-\lambda)\right] =$$

$$= (3-\lambda)^2(4-\lambda) - 2(3-\lambda) =$$

$$= (3-\lambda)\left((3-\lambda)(4-\lambda) - 2\right) = (3-\lambda)\left(12 - 3\lambda - 4\lambda + \lambda^2 - 2\right) =$$

$$= (3-\lambda)\left(\lambda^2 - 7\lambda + 10\right) = (3-\lambda)(\lambda-2)(\lambda-5)$$

$\lambda_1 = 2$, $\lambda_2 = 3$, $\lambda_3 = 5$

$$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix} \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & -1 & 2 & 0 \end{array}\right] \quad R_3 \leftarrow R_3 - R_1 \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \end{array}\right]$$

$$R_3 \leftarrow R_3 + R_2 \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \begin{array}{l} z = t \\ x = -t \\ y = t \end{array} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = v_1$$

$$A - 3I = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix} \Rightarrow \left[\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 1 & -1 & 1 & 0 \end{array}\right] \quad R_1 \leftrightarrow R_3 \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$$\begin{array}{l} R_1 \leftarrow R_1 + R_2 \\ R_2 \leftarrow R_2 + R_3 \end{array} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \begin{array}{l} y = t \\ x = t \\ z = 0 \end{array} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = v_2$$

$$A - 5I = \begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix} \quad R_1 \leftrightarrow R_3 \left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & -2 & -1 & 0 \\ -2 & 0 & 1 & 0 \end{array}\right] \quad R_3 \leftarrow R_3 + 2R_1 \left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & -2 & -1 & 0 \end{array}\right]$$

15

Problem 6. [5'] Find a, b, and c such that $\mathrm{w} = a\mathrm{v}_1 + b\mathrm{v}_2 + c\mathrm{v}_3$, where

$$\mathrm{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathrm{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathrm{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathrm{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right] \quad R_3 \leftarrow R_3 + R_2 \quad \left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ 0 & 1 & 1 & 0 \end{array}\right] \quad R_1 \leftarrow R_1 - 2R_3$$

$$\left[\begin{array}{ccc|c} 0 & 0 & -3 & -3 \\ 2 & -1 & 1 & 8 \\ 0 & 1 & 1 & 0 \end{array}\right] \quad \begin{array}{l} R_1 \leftarrow -\frac{1}{3}R_1 \\ R_2 \leftarrow R_2 + R_3 \end{array} \quad \left[\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 2 & 0 & 2 & 8 \\ 0 & 1 & 1 & 0 \end{array}\right] \quad \begin{array}{l} R_3 \leftarrow R_3 - R_1 \\ R_2 \leftarrow \frac{1}{2}R_2 \end{array}$$

$$\left[\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 4 \\ 0 & 1 & 0 & -1 \end{array}\right] \quad R_2 \leftarrow R_2 - R_1 \quad \left[\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \end{array}\right] \quad \begin{array}{l} R_1 \leftrightarrow R_2 \\ R_2 \leftrightarrow R_3 \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$a = 3, \quad b = -1, \quad c = 1$

$\mathrm{w} = 3\mathrm{v}_1 - \mathrm{v}_2 + \mathrm{v}_3$

$$\begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix} = 3\begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix} - \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Problem 7. [5] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.

$v_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$, $v_2 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}$

orthogonal:

$|u_1| = \sqrt{0^2+1^2+1^2+0^2} = \sqrt{2}$, $|u_1|^2 = 2$

$u_1 = v_1 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$

$u_1 \cdot v_2 = \begin{bmatrix}0\\1\\1\\0\end{bmatrix} \cdot \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} = 0+0+1\cdot 2+0(-1) = 2$

$u_2 = v_2 - \frac{u_1 \cdot v_2}{|u_1|^2} u_1 = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \frac{2}{2} \cdot \begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}3\\0\\2\\-1\end{bmatrix} - \begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix}$

orthonormal:

$\frac{u_1}{|u_1|} = \frac{1}{\sqrt{2}} \cdot \begin{bmatrix}0\\1\\1\\0\end{bmatrix} = \begin{bmatrix}0\\1/\sqrt{2}\\1/\sqrt{2}\\0\end{bmatrix}$

$|u_2| = \sqrt{3^2+(-1)^2+1^2+(-1)^2} = \sqrt{9+1+1+1} = \sqrt{12}$

$\frac{u_2}{|u_2|} = \frac{1}{\sqrt{12}} \begin{bmatrix}3\\-1\\1\\-1\end{bmatrix} = \begin{bmatrix}3/\sqrt{12}\\-1/\sqrt{12}\\1/\sqrt{12}\\-1/\sqrt{12}\end{bmatrix}$

orthonormal basis: $\left\{\begin{bmatrix}0\\1/\sqrt{2}\\1/\sqrt{2}\\0\end{bmatrix}, \begin{bmatrix}3/\sqrt{12}\\-1/\sqrt{12}\\1/\sqrt{12}\\-1/\sqrt{12}\end{bmatrix}\right\}$

1.5/5

Problem 8. [5'] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

If $A^T = -A \Rightarrow$ A is antisymmetric

n is odd. Let's take $n=3$ for example

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \quad A^T = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix} = -A \Rightarrow$$

$a = -a$, $a = 0$
$b = -d$
$c = -g$
$e = -e$, $e = 0$
$f = -h$
$i = -i$, $i = 0$

$a = e = i = 0 \Rightarrow$ main diagonal of A has all zeroes

$$A^T = \begin{bmatrix} 0 & -d & -g \\ d & 0 & -h \\ g & h & 0 \end{bmatrix}, \quad \det(A^T) = 0 \cdot 0 \cdot 0 + (-d)(-h)g + (-g)dh = 0 + dhg - dhg = 0$$

You can't prove by example

$\det(A) = \det(A^T) = 0$

This is true for all odd n:

The entry a in position (i,i) in matrix A is in position (i,i) in A^T too, and since A is antisymmetric $a = -a \Rightarrow a = 0$ The main diagonal contains all zeroes.

For other entries in position (i,j), in A^T they will be in position (j,i). $a_{ij} = -a_{ji}$. When you calculate the determinant these entries will cancel out, giving $\det(A^T) = 0$. Since $\det(A^T) = \det(A)$, $\det(A) = 0$.

0.5/5

Problem 9. [5] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

$u \cdot v = 0$

For triangle inequality : for $u, v \in \mathbb{R}^n$:

$\|u+v\| \leq \|u\| + \|v\|$

$\|u-v\| \leq \|u\| - \|v\|$

No

if $\|u+v\| = \|u-v\|$, $\|u\|+\|v\| = \|u\| - \|v\|$

$2\|v\| = 0 \Rightarrow \|v\| = 0$

$\Rightarrow v = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$

$u \cdot v = u \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} = 0 \Rightarrow u \cdot v = 0, \; u \perp v$

u and v are perpendicular to each other